

TCS NQT Preparation Guide

Aptitude, Advanced Aptitude, DSA concepts, and practice problems - placement focused

How to use this PDF: First finish basic aptitude and SQL/DBMS foundations. Then practice advanced aptitude and DSA daily. For TCS NQT, focus on speed, accuracy, and repeated revision.

1. TCS NQT Preparation Roadmap

Stage	What to study	Daily target
Stage 1 - Basics	Number system, percentages, ratios, averages, time and work, profit and loss, simple SQL, basic programming logic	2 hours aptitude + 1 hour coding
Stage 2 - Speed	Arithmetic shortcuts, mental calculation, verbal ability, reasoning, MCQ practice	40-60 questions daily
Stage 3 - Advanced	Probability, permutations and combinations, functions, logarithms, advanced reasoning, coding MCQs	25 advanced questions daily
Stage 4 - DSA	Arrays, strings, searching, sorting, stack, queue, linked list, recursion, trees, hash maps	2 coding problems daily
Stage 5 - Mock tests	Full-length TCS NQT mock tests with timer and analysis	1 mock test every 2-3 days

- Do not only watch tutorials. Practice questions immediately after each topic.
- Maintain an error notebook: formula mistakes, concept mistakes, and silly mistakes.
- For coding, solve easy problems first, then medium problems. Do not jump directly to hard problems.

2. TCS NQT Basic Aptitude Topics

Area	Topics to learn	Practice focus
Number System	LCM, HCF, divisibility rules, remainders, unit digit, last two digits, factors, prime numbers	Fast calculation and pattern-based questions
Simplification	BODMAS, fractions, decimals, surds, indices, approximations	Speed and accuracy
Percentages	Percentage change, successive percentage, marks, population, price changes	Conversion between fraction and percentage
Profit and Loss	Cost price, selling price, discount, marked price, gain/loss percentage	Successive discounts and dishonest shopkeeper basics
Simple and Compound Interest	SI, CI, difference between SI and CI, annual/half-yearly compounding	Formula application and comparison
Ratio and Proportion	Basic ratio, compound ratio, partnership, ages, mixtures	Form equations quickly
Average	Basic average, weighted average, average speed, replacement questions	Data interpretation style questions
Time and Work	Work efficiency, pipes and cisterns, wages, combined work	LCM method and efficiency method
Time, Speed, Distance	Relative speed, trains, boats and streams, races, circular tracks	Unit conversion and relative motion
Mensuration	Area, perimeter, volume, surface area of basic shapes	Formula memory and diagram understanding
Data Interpretation	Tables, bar graphs, pie charts, line graphs, caselets	Percentage and ratio interpretation

3. TCS NQT Reasoning Topics

Area	Topics	Practice focus
Verbal Reasoning	Analogy, classification, coding-decoding, blood relations, direction sense	Understand relation patterns
Non-Verbal Reasoning	Series, mirror images, embedded figures, paper folding, figure counting	Visual pattern observation
Logical Reasoning	Syllogism, statement and conclusion, assumptions, arguments, course of action	Read carefully; avoid personal assumptions
Arrangement	Seating arrangement, linear/circular arrangement, ranking, order sequence	Draw clean diagrams
Puzzles	Box puzzles, scheduling, floor puzzles, comparison puzzles	Use tables and eliminate cases
Data Sufficiency	Check whether given statements are enough to answer	Do not solve fully unless needed

4. TCS NQT Verbal Ability Topics

Area	Topics	Practice focus
Grammar	Parts of speech, tenses, subject-verb agreement, articles, prepositions, conjunctions	Error spotting and sentence correction
Vocabulary	Synonyms, antonyms, one-word substitutions, idioms and phrases	Learn through examples, not only meanings
Reading	Reading comprehension, central idea, inference, tone, title	Read question first, then passage
Sentence Skills	Para jumbles, sentence completion, fill in the blanks, cloze test	Logical flow and grammar clues
Email/Business English	Formal tone, concise writing, professional vocabulary	Placement communication rounds

5. Advanced Aptitude Topics for TCS NQT

Area	Topics to learn	Why important
Permutation and Combination	Factorial, arrangements, selections, circular arrangement, repeated objects	Appears in advanced quantitative questions
Probability	Basic probability, independent events, mutually exclusive events, conditional probability basics	Common in advanced aptitude and coding logic
Algebra	Linear equations, quadratic equations, inequalities, functions, graphs	Useful for problem solving and MCQs
Logarithms and Exponents	Log rules, exponential equations, growth comparison	Needed for advanced arithmetic and CS concepts
Set Theory	Venn diagrams, union, intersection, complement, inclusion-exclusion	Useful in aptitude and discrete mathematics
Progressions	AP, GP, HP basics, nth term, sum of series	Frequently asked in pattern-based questions
Advanced DI	Mixed charts, missing data, caselets, comparison-based DI	Tests speed and decision making
Advanced Geometry	Triangles, circles, coordinate geometry basics	Useful for mensuration and reasoning
Statistics	Mean, median, mode, variance basics, standard deviation basics	Asked in data interpretation and analytics roles
Advanced Reasoning	Complex puzzles, input-output, decision making, critical reasoning	High scoring if practiced properly

6. Important Aptitude Formulas and Shortcuts

Topic	Must remember
Percentage	$x\% \text{ of } y = y\% \text{ of } x$. Successive change: $a + b + \frac{ab}{100}$.
Profit/Loss	$\text{Profit}\% = \frac{\text{Profit}}{\text{CP}} \times 100$. $\text{Loss}\% = \frac{\text{Loss}}{\text{CP}} \times 100$. $\text{Discount}\% = \frac{\text{Discount}}{\text{MP}} \times 100$.
SI/CI	$\text{SI} = \frac{\text{PRT}}{100}$. CI amount = $P(1 + \frac{R}{100})^T$.
Average	Average = $\frac{\text{sum of observations}}{\text{number of observations}}$. Weighted average = $\frac{\text{total weighted value}}{\text{total weight}}$.
Time and Work	If A finishes in x days, one day work = $\frac{1}{x}$. Combined work = sum of individual one day works.
Speed	Speed = $\frac{\text{distance}}{\text{time}}$. $1 \text{ m/s} = \frac{18}{5} \text{ km/h}$. $1 \text{ km/h} = \frac{5}{18} \text{ m/s}$.
Permutation	$nPr = \frac{n!}{(n-r)!}$. Use when order matters.
Combination	$nCr = \frac{n!}{r!(n-r)!}$. Use when order does not matter.
Probability	Probability = $\frac{\text{favourable outcomes}}{\text{total outcomes}}$.
Set Theory	$n(A \cup B) = n(A) + n(B) - n(A \cap B)$.

7. DSA Concepts to Learn for TCS NQT and Placements

DSA concept	What to learn	Must practice
Time and Space Complexity	Big O, Big Omega, Big Theta, best/average/worst case	Analyze loops, nested loops, recursion
Arrays	Indexing, traversal, prefix sum, sliding window, two pointers	Max/min, subarray, rotation, duplicates
Strings	Character frequency, palindrome, anagram, substring, pattern matching basics	Reverse, count vowels, anagram, first unique character
Searching	Linear search, binary search, lower bound, upper bound	Search in sorted array, first/last occurrence
Sorting	Bubble, selection, insertion, merge, quick sort concepts	Sort numbers, sort strings, custom sorting basics
Hashing	Hash map, hash set, frequency table	Two sum, duplicates, frequency count
Stack	LIFO, push, pop, peek, balanced brackets	Parentheses, next greater element
Queue	FIFO, enqueue, dequeue, circular queue, deque basics	Queue simulation, first non-repeating character
Linked List	Node, head, traversal, insert, delete, reverse	Find middle, detect cycle, merge two lists
Recursion	Base case, recursive case, call stack	Factorial, Fibonacci, sum of digits, recursion on arrays
Trees	Binary tree, BST, traversal: inorder, preorder, postorder, level order	Height, search in BST, traversal output
Graphs	Vertices, edges, BFS, DFS, adjacency list/matrix	Connected components, path existence basics
Dynamic Programming Basics	Overlapping subproblems, memoization, tabulation	Fibonacci, climbing stairs, knapsack basics

8. Programming Foundations Before DSA

- Variables, data types, input and output, operators, type conversion.
- Conditions: if, else if, else, switch/case if your language supports it.
- Loops: for, while, nested loops, break, continue.
- Functions: parameters, return values, scope, reusable logic.
- Arrays/lists and strings: indexing, traversal, update, searching.
- Basic debugging: print intermediate values and test with small input first.

9. DSA Practice Problems - Beginner Level

No.	Problem	Concept
1	Find maximum and minimum element in an array.	Array traversal
2	Reverse an array without using extra array.	Two pointers
3	Find sum and average of array elements.	Array basics
4	Count even and odd numbers in an array.	Conditions + array
5	Find second largest element in an array.	Array logic
6	Check whether a number is prime.	Loops
7	Print Fibonacci series up to n terms.	Loops/recursion
8	Find factorial of a number.	Loop/recursion
9	Reverse a string.	String/two pointers
10	Check whether a string is palindrome.	String
11	Count vowels, consonants, digits, and spaces in a string.	String traversal
12	Find frequency of each character in a string.	Hashing
13	Remove duplicate elements from an array.	Hash set
14	Check if two strings are anagrams.	Hashing/string
15	Linear search in an array.	Searching
16	Binary search in a sorted array.	Searching
17	Sort an array using bubble sort.	Sorting
18	Sort an array using insertion sort.	Sorting
19	Implement stack using array/list.	Stack
20	Check balanced parentheses.	Stack

10. DSA Practice Problems - Intermediate Level

No.	Problem	Concept
1	Find pair with given sum in an array.	Hashing/two pointers
2	Move all zeros to the end while maintaining order.	Two pointers
3	Rotate array by k positions.	Array manipulation
4	Find missing number from 1 to n.	Math/XOR
5	Find duplicate number in an array.	Hashing/Floyd cycle
6	Find first non-repeating character in a string.	Hashing/queue
7	Find longest substring without repeating characters.	Sliding window
8	Find maximum sum subarray.	Kadane algorithm
9	Merge two sorted arrays.	Two pointers
10	Find first and last position of element in sorted array.	Binary search
11	Find next greater element for each element.	Stack
12	Implement queue using two stacks.	Stack/queue
13	Reverse a linked list.	Linked list
14	Find middle node of a linked list.	Slow-fast pointer

No.	Problem	Concept
15	Detect cycle in a linked list.	Slow-fast pointer
16	Merge two sorted linked lists.	Linked list
17	Print inorder, preorder, and postorder traversal.	Tree traversal
18	Find height of a binary tree.	Tree recursion
19	Search an element in BST.	BST
20	BFS and DFS traversal of a graph.	Graph basics

11. DSA Practice Problems - TCS Style Coding

No.	Problem statement	Expected concept
1	Given a number n, find the sum of digits until a single digit is obtained.	Loops/math
2	Given an integer, check whether it is an Armstrong number.	Math
3	Given a string, reverse words in the string.	String
4	Given an array, find the leader elements. An element is leader if all elements to its right are smaller.	Array traversal
5	Given a sorted array, remove duplicates in-place.	Two pointers
6	Given an array, count frequency of each element and print in sorted order.	Hashing/sorting
7	Given two strings, check whether one is rotation of the other.	String
8	Given n, print all prime numbers up to n.	Sieve/basic loops
9	Given an array and k, find maximum sum of subarray of size k.	Sliding window
10	Given a matrix, print elements in spiral order.	Matrix
11	Convert decimal number to binary and binary to decimal.	Number system
12	Find GCD and LCM of two numbers.	Euclidean algorithm
13	Given a sentence, find the longest word.	String
14	Given an array, find equilibrium index where left sum equals right sum.	Prefix sum
15	Given a stack, sort it using another stack.	Stack

12. Aptitude Practice Questions

- 1. A number is increased by 20% and then decreased by 20%. What is the net percentage change?
- 2. A shopkeeper gives two successive discounts of 10% and 20%. What is the equivalent single discount?
- 3. A can complete a work in 12 days and B can complete it in 18 days. In how many days can they complete it together?
- 4. A train 120 m long crosses a pole in 6 seconds. Find its speed in km/h.
- 5. The average of 5 numbers is 24. If one number 30 is removed, what is the new average if the remaining sum is 90?
- 6. In how many ways can 3 students be selected from 8 students?
- 7. A coin is tossed twice. What is the probability of getting exactly one head?
- 8. If the ratio of ages of A and B is 3:4 and their total age is 35, find their ages.
- 9. Find the next number: 2, 6, 12, 20, 30, ?
- 10. If $n(A)=40$, $n(B)=30$, and $n(A \cap B)=10$, find $n(A \cup B)$.

13. Answers for Aptitude Practice

Q	Answer
1	4% decrease. Formula: $a + b + ab/100 = 20 - 20 - 4 = -4\%$.
2	28% discount. Formula: $10 + 20 - (10 \times 20)/100 = 28\%$.
3	$36/5 = 7.2$ days. Combined work = $1/12 + 1/18 = 5/36$.
4	72 km/h. Speed = $120/6 = 20$ m/s = $20 \times 18/5 = 72$ km/h.
5	22.5. New average = $90/4$.
6	56. $8C3 = 56$.
7	$1/2$. Outcomes: HH, HT, TH, TT. Exactly one head = 2 outcomes.

Q	Answer
8	A = 15 years, B = 20 years. Total parts = 7, one part = 5.
9	42. Differences are 4, 6, 8, 10, so next difference is 12.
10	60. Union = $40 + 30 - 10$.

14. 30-Day Study Plan

Days	Aptitude/Reasoning	DSA/Coding
1-5	Number system, simplification, percentages, ratios	Programming basics, arrays, strings
6-10	Profit/loss, SI/CI, average, time and work	Searching, sorting, hashing
11-15	Time-speed-distance, mensuration, DI basics	Stack, queue, recursion
16-20	P&C, probability, set theory, progressions	Linked list, two pointers, sliding window
21-25	Advanced reasoning, puzzles, data sufficiency	Trees, BST, graph basics
26-30	Mock tests, revision, error notebook, weak topics	TCS-style coding problems and revision

15. Best Practice Method

- For each topic: learn concept -> solve 10 easy questions -> solve 10 timed questions -> revise mistakes.
- For aptitude: focus on accuracy first, then speed. Speed comes after repeated practice.
- For DSA: write code yourself. Do not only read solutions. Dry run with sample input.
- Before the exam: revise formulas, solve mock tests, and avoid learning completely new topics at the last minute.
- For TCS NQT coding: practice simple and clean solutions. Use meaningful variable names and handle edge cases.

Final advice: Bro, for TCS NQT your target should be: basic aptitude strong, advanced aptitude decent, DSA basics strong, and regular mock practice. Consistency matters more than learning too many topics randomly.